



What "Training a Model" Actually Means

Pretraining · Fine-tuning · RLHF · Distillation: four completely different processes, one overloaded word.

1. Pretraining: Builds foundation knowledge (\$10M+, months)

2. Fine-tuning (SFT): Shapes instruction-following behavior (\$50, hours)

3. RLHF / DPO: Aligns human preferences & safety (weeks)

4. Distillation: Compresses large teacher into small student (days)



The Confusion This Solves

"We trained a model" can mean four very different things

When someone says they "trained" a model, they might mean:

-  **Pretraining:** Millions of dollars over months on thousands of GPUs.
-  **Fine-tuning:** Running a script on 500 examples for 20 minutes (\$50).
-  **RLHF / DPO:** Aligning behavior using human preference rankings.
-  **Distillation:** Compressing a large model into a small one.

Disambiguation Branching Diagram

- | | |
|-------------|------------------------------|
| • Pretrain | \$10M+ Months 1000s GPUs |
| • Fine-tune | \$50 Hours 1 GPU |
| • RLHF/DPO | Varies Weeks Human Taste |
| • Distill | Varies Days 70B → 7B |



Using the right word: and knowing which one you actually need: saves enormous time and money.



The Standard Pipeline

How a modern chat model actually gets built

1. PRETRAINING

Predicts text continuations (not a chatbot yet).

knows a lot, useless to talk to

2. SUPERVISED FINE-TUNING (SFT)

Teaches the model to follow instructions and answer Q&A.

3. PREFERENCE TUNING (RLHF / DPO)

Makes the model helpful, safe, honest, and well-behaved.

4. (OPTIONAL) DISTILLATION

Transfers capability into a smaller, faster model.

Artifact Transformation Flowchart

[Raw Web Corpus (Trillions)] → **Stage 1** → [Base Checkpoint]

[Instruction Pairs (10k)] → **Stage 2** → [Instruct Model]

[Preference Pairs A vs B] → **Stage 3** → [Aligned Chatbot]

[Teacher Synthesized Data] → **Stage 4** → [7B Edge Model]



Key Insight: A freshly pretrained model is NOT a chatbot. Everything after step 1 shapes *behavior*, not knowledge.



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Stage 1: Pretraining

Learning language itself, from scratch

The model starts with random weights and learns by predicting the next token across an enormous corpus: trillions of tokens of web text, books, and code.

Input: "The capital of France is"

Target: " Paris"

↓

adjust billions of weights slightly → repeat trillions of times

Self-Supervised Next-Token Training Loop

[Raw Corpus] → [Tokenizer] → [Forward Pass] → [Cross-Entropy Loss] →
[AdamW Weight Update]

★ What makes pretraining distinctive:

✓ **Self-supervised:** No human labels needed; text provides answer.

\$ **Enormously expensive:** 1000s of GPUs, months, \$millions.

🧠 **Where knowledge comes from:** Facts, grammar, code absorbed here.

🏢 **Frontier Labs only:** Essentially never done by individual teams.



⚠ BASE MODEL LIMITATION

Base Model Behavior

Why pretraining alone isn't enough

A pretrained "base model" completes text: it doesn't answer questions. That distinction surprises people.

Prompt to BASE Model:

"What is the capital of France?"

Likely Output:

"What is the capital of Germany? What is the capital of Spain?
What is the largest city in Europe? ... "

↑ continuing a plausible list of questions, not answering!

Behavior Across Prompt Types

Prompt	Base Model	Instruct Model
Question	More questions	Direct Answer
Code	Completes file	Explains & implements

Why this happens: Training data contains countless quiz lists. Statistically, more questions is a reasonable continuation. The model has no concept of "being helpful" until later stages.



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Stage 2: Fine-Tuning (SFT)

Teaching the model to follow instructions

Continue training the pretrained model, but now on curated **instruction-response pairs** written or vetted by humans.

```
{ "messages": [  
  { "role": "user", "content": "Explain photosynthesis  
simply." } ,  
  { "role": "assistant", "content": "Plants use sunlight to  
turn water ... " }  
]}
```

SFT Target Token Loss Masking

User Tokens: [Explain ...] → **Loss Masked (0.0)**

Assistant Tokens: [Plants ...] → **Loss Calculated (1.0)**

What SFT changes:

- ✓ **Learns Assistant Format:** Question gets an answer, not continuation.
- ✓ **Learns Tone & Style:** Structure, tone, and formatting constraints.
- ✗ **No New Knowledge:** Does NOT add facts: those came from pretraining.



Domain Fine-Tuning

The kind of fine-tuning realistic for your team

Taking an existing instruct model and further tuning it on your domain's data: the version of "training" most engineers will actually run.

✓ When it helps:

- Consistent format/structure
- Domain tone & terminology
- Reduce long prompt headers
- High-volume narrow task

✗ When it fails:

- **Adding facts** (Use RAG instead!)
- Anything unsolved by prompting

🏗 Architectural Decision Diagram

FINE-TUNING

Bakes behavior into weights
"HOW to respond"

RAG VECTOR SEARCH

Injects dynamic context
"WHAT to know"



RAG is 10x cheaper and provides real-time updateability without retraining risk!



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Stage 3: RLHF

Reinforcement Learning from Human Feedback

SFT teaches the model to imitate good answers. RLHF goes further: it teaches the model what humans **prefer** when several plausible answers exist.

🔄 The 3-Step RLHF Loop

- 1. Collect Preferences:** Show humans 2 model outputs for same prompt → pick better one.
- 2. Train Reward Model:** A separate model learns to predict human preference scores.
- 3. Optimize LLM (PPO):** Adjust LLM so it generates outputs scoring high on reward model.

🔗 3-Model PPO System Architecture

[Actor LLM] → Output → [Reward Model Score] - (KL Penalty vs [Ref Model]) → PPO Loss

❤️ Turns human taste and judgment into an optimizable reward function.



Why RLHF Matters

The stage that made chat models actually usable

✓ What it achieves:

- Genuinely helpful answers
- Safety & harmful refusals
- Honest uncertainty tone
- Captures human taste

⚠ Difficulties:

- Reward hacking (verbose)
- Costly human labeling
- Unstable PPO training
- Encodes annotator bias

🚫 Reward Hacking Failure Mode

Models learn to write 500-word verbose essays with endless disclaimers because human annotators systematically rate longer answers higher, even when less accurate!



RLHF is why models refuse dangerous prompts and maintain polite conversation styles.



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DPO: Modern Alternative

Direct Preference Optimization: same goal, far less machinery

DPO achieves preference alignment **without** training a separate reward model or running reinforcement learning at all.

🔗 Architectural Complexity Comparison

RLHF: Prefs → [Reward Model] → PPO RL → [Aligned Model] (4 models)

DPO: Prefs → [Direct Implicit Loss] → [Aligned Model] (1 model!)

✅ Why DPO is widely used:

- Simpler to implement & far more stable to train.
- Uses same preference pairs (chosen vs rejected response).
- Comparable alignment at lower complexity and cost.

Related variants: ORPO, KTO, and **RLAIF** (AI Feedback: an LLM provides preference labels instead of humans).



Stage 4: Distillation

Teaching a small model to imitate a large one

A large, capable **teacher** model generates outputs; a smaller **student** model is trained to reproduce them: inheriting capability at a fraction of the size.

📥 Knowledge Transfer Pipeline

TEACHER (70B) → **Logits / Outputs** → STUDENT (7B)

Response Distillation: Student trains on teacher's final output text (simple API access).

Logit Distillation: Student matches full probability distributions (needs internal access).

🕒 **Efficiency Impact:** 7B student runs 10x faster and requires 10x less GPU RAM than 70B teacher!





Why Distillation Matters

Where most small models you use come from

- ✓ **Small models punch high:** Nearly every strong 3B-7B model is distilled from a larger model.
- ✓ **Reasoning transfer:** Large model reasoning traces teach small models to think step-by-step.
- ! **Inherits mistakes:** Student inherits teacher's capabilities AND biases/errors.
- 🔪 **Legal Caution:** Many commercial API Terms of Service prohibit training competitor models from outputs.

🧠 Case Study: DeepSeek-R1 Distillation Pipeline

[DeepSeek-R1 671B] → 800k Reasoning Traces → [SFT on Qwen 1.5B/7B/14B] → SOTA Small Models



Distillation is the key engine powering efficient local edge models.



LoRA & QLoRA

How fine-tuning became affordable

Full fine-tuning updates every weight: requiring massive GPU RAM. **LoRA** freezes base weights and trains small "adapter" matrices instead.

Low-Rank Matrix Decomposition Math

$$W_{\text{new}} = [\text{Frozen Base } W_0 \text{ (7B)}] + [\Delta W = \text{Matrix A} \times \text{Matrix B} \text{ (~0.1\% params)}]$$

```
from peft import LoraConfig, get_peft_model
config = LoraConfig(r=8, lora_alpha=16, target_modules=
["q_proj", "v_proj"])
model = get_peft_model(base_model, config)
model.print_trainable_parameters() # "trainable: 0.12% of all
params"
```

 **VRAM Savings:** Full 70B tuning requires 140GB+ VRAM (8x A100s). QLoRA 70B requires only 48GB VRAM (Single GPU)!



Full Comparison

All four processes, side by side

Feature	Pretraining	SFT	RLHF/DPO	Distillation
Goal	Knowledge	Instructions	Preferences	Compress
Data	Trillions tokens	Q&A pairs	Rankings A/B	Teacher outputs
Cost	\$Millions	\$10s-\$100s	Moderate	Moderate
Time	Months	Hours	Weeks	Days
Changes	Everything	Format/Tone	Safety/Style	Model Size
Who does it	Frontier Labs	Anyone	Labs/Teams	Labs/Teams



Bookmark this matrix: use it when planning your team's LLM customization strategy.



Which Do You Need?

A decision guide for your own project

Project Decision Flowchart

Need FACTS? → Use RAG (Not Training)

Need FORMAT / STYLE? → Supervised Fine-Tuning (LoRA)

Need CHEAP EDGE MODEL? → Distillation

Need ALIGNMENT / SAFETY? → DPO



Cost-Saving Ladder: Always try Prompting → RAG → Fine-Tuning in that exact order!



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⚠ MYTH BUSTING

Common Misconceptions

What people get wrong about training

✗ ~~Fine-tuning adds facts~~

Unreliable for facts. Use RAG.

✗ ~~More data is better~~

High quality beats noisy pairs.

✗ ~~RLHF = Fine-Tuning~~

SFT imitates; RLHF optimizes.

✗ ~~Distill = Compression~~

Student is a separate model.

✗ ~~You must fine-tune~~

Prompting + RAG covers 90% apps.

✗ ~~Forgetting isn't real~~

Aggressive tuning degrades skills.



Avoid costly traps by starting with simple retrieval & clear prompts before fine-tuning!



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🎯 CORE TAKEAWAY

Four processes, one word

Pretraining builds knowledge. Fine-tuning shapes behavior.
RLHF/DPO aligns preferences. Distillation compresses size.

PRETRAIN → Knowledge

FINE-TUNE → Behavior & Format

RLHF / DPO → Preferences & Safety

DISTILL → Size & Speed Efficiency

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